

F2 - Buckets

分值: 300 分

Problem Statement

Problems F and F2 are the same problem with different constraints on M . In Problem F2, $M = 5$.

题目F和F2是同一道题，仅对 M 的约束条件不同。在F2中， $M = 5$ 。

You are given a positive integer M . Consider the following problem.

给定一个正整数 M 。请考虑以下问题。

Bucket

You are given a positive integer N and non-negative integer sequences $A = (A_1, A_2, \dots, A_N)$ and $B = (B_1, B_2, \dots, B_N)$ of length N . All elements of A and B are between 0 and M , inclusive.

There are N buckets numbered 1 to N . Each bucket can hold up to M liters of water. Initially, bucket i contains A_i liters of water.

You may perform the following operation any number of times.

- Choose two distinct buckets i and j . Continue pouring water from bucket i into bucket j as long as both of the following conditions are satisfied:
 - Bucket i still has water remaining.
 - The amount of water in bucket j is less than M liters.

Your goal is to have exactly B_i liters of water in bucket i for all i . Determine whether the goal can be achieved.

桶问题

给定正整数 N ，以及长度为 N 的非负整数序列 $A = (A_1, A_2, \dots, A_N)$ 和 $B = (B_1, B_2, \dots, B_N)$ 。 A 和 B 的所有元素均在0到 M 之间（含两端）。

共有 N 个桶，编号为1到 N 。每个桶最多可装 M 升水。初始时，桶 i 中有 A_i 升水。

你可以执行任意次以下操作：

- 选择两个不同的桶 i 和 j 。只要同时满足以下两个条件，就持续将桶 i 中的水倒入桶 j ：
 - 桶 i 中仍有剩余水。
 - 桶 j 中的水量小于 M 升。

你的目标是让所有 i 对应的桶 i 中恰好有 B_i 升水。判断能否达成目标。

You are given a non-negative integer matrix $X = (X_{i,j}) (0 \leq i, j \leq M)$ of $(M + 1)$ rows and $(M + 1)$ columns. Process the following queries Q times.

给定一个大小为 $(M + 1)$ 行 $(M + 1)$ 列的非负整数矩阵 $X = (X_{i,j}) (0 \leq i, j \leq M)$ 。请处理共 Q 次查询。

- You are given non-negative integers i, j, Y with $0 \leq i, j \leq M$. Change $X_{i,j}$ to Y . Then, obtain non-negative integer sequences A and B by the following procedure.
 - Initialize non-negative integer sequences A and B as empty sequences.
 - For $i = 0, 1, \dots, M$ in this order:
- 给定满足 $0 \leq i, j \leq M$ 的非负整数 i, j, Y 。将 $X_{i,j}$ 修改为 Y 。随后通过以下步骤得到非负整数序列 A 和 B ：
 - 初始化非负整数序列 A 和 B 为空序列。
 - 按顺序对 $i = 0, 1, \dots, M$ 执行操作：

- For $j = 0, 1, \dots, M$ in this order:
- 按顺序对 $j = 0, 1, \dots, M$ 执行操作：

- Do this $X_{i,j}$ times: append i to the end of A and j to the end of B .
- 重复此操作 $X_{i,j}$ 次：将 i 追加到 A 的末尾，将 j 追加到 B 的末尾。

- Solve **Bucket** for the non-negative integer sequences A and B of length $N = \sum_{i=0}^M \sum_{j=0}^M X_{i,j}$.
- 针对长度为 $N = \sum_{i=0}^M \sum_{j=0}^M X_{i,j}$ 的非负整数序列 A 和 B ，求解上述桶问题。

Constraints

- $M = 5$
- $1 \leq Q \leq 10^6$
- $0 \leq X_{i,j}, Y \leq 10^{17}$
- $0 \leq i, j \leq M$

- $\sum_{i=0}^M \sum_{j=0}^M X_{i,j} \geq 1$ at any time.
- All input values are integers.
- $M = 5$
- $1 \leq Q \leq 10^6$
- $0 \leq X_{i,j}, Y \leq 10^{17}$
- $0 \leq i, j \leq M$
- 任意时刻的 $\sum_{i=0}^M \sum_{j=0}^M X_{i,j} \geq 1$.
- 所有输入值均为整数。

Input

The input is given from Standard Input in the following format:

输入从标准输入按以下格式给出：

```

M Q
X0,0 X0,1 ... X0,M
X1,0 X1,1 ... X1,M
⋮
XM,0 XM,1 ... XM,M
query1
query2
⋮
queryQ

```

Each query is given in the following format:

每个查询按以下格式给出：

```

i j Y

```

Output

Output Q lines. The i -th line should contain Yes if the goal can be achieved in query _{i} , and No otherwise.

输出 Q 行。第 i 行中，如果第query _{i} 次查询的目标可以达成，输出Yes，否则输出No。

Sample Input 1

```
5 3
1 0 0 0 0 0
0 0 0 0 0 0
0 0 0 0 0 0
0 1 0 0 0 1
0 0 0 0 0 0
0 0 0 0 0 0
0 5 0
4 0 1
1 5 1
```

Sample Output 1

```
Yes
No
Yes
```

The sequences A and B obtained at each query are as follows:

每次查询得到的序列 A 和 B 如下:

- At the time of processing the first query: $A = (0, 3, 3), B = (0, 1, 5)$
- At the time of processing the second query: $A = (0, 3, 3, 4), B = (0, 1, 5, 0)$
- At the time of processing the third query: $A = (0, 1, 3, 3, 4), B = (0, 5, 1, 5, 0)$
- 处理第一次查询时: $A = (0, 3, 3), B = (0, 1, 5)$
- 处理第二次查询时: $A = (0, 3, 3, 4), B = (0, 1, 5, 0)$
- 处理第三次查询时: $A = (0, 1, 3, 3, 4), B = (0, 5, 1, 5, 0)$

Sample Input 2

```
5 10
61717749303507807 83849626163233111 84388501405824055 8514730661576685 8408052512771
```

33333333333333333333 70477132070314067 43546706335438313 61370380435458585 991018236896
33333333333333333333 21459633523757247 8151410984806542 14183403125185219 6829918611068
98146856836553017 53630682913685434 12400422817799555 29967281381593348 675215471364
97356055491517734 55777507072845580 59235925940735000 6228770338558507 417781086082
5960356361755846 59147828067221624 65687376011190687 33333333333333333333 3156384883225
2 3 1333175914347793
3 4 80371774347266293
2 1 20920094361040152
2 3 793636751630698
0 5 68209364995103050
2 0 98643963346205063
3 2 7931291059115265
0 1 79380494404548821
2 5 6440038070576840
2 3 88122600737306767

Sample Output 2

No
No
No
No
No
Yes
No
Yes
No
Yes